NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 1

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1. Convert each of the following decimal numbers to unsigned binary, octal and hexadecimal numbers: (a) 27, (b) 915.

   Ans: (a) 11011_2, 33_8, & $1B_{16}$
   (b) $0011\ 1001\ 0011_2$, 1623_8, & 393_{16}

2. Convert the following numbers into decimal: (a) 4310_5, (b) 198_{12}, (c) EEE_{16}.

   Ans: (a) 580_{10} (b) 260_{10} (c) 3822_{10}

3. Determine the base of the numbers in each case for the following operations to be correct: (a) 14/2 = 5, (b) 23 × 14 = 333, (c) 24+17 = 40.

   Ans: (a) 6 (b) 9 (c) 11

4. Without converting to decimal, perform addition, subtraction and multiplication of the two numbers, 634_7 and 253_7. Show all your workings.

   Ans: 1220_7, 351_7 and 240325_7

5. Obtain the decimal values of the following 2's complement numbers. From the results, what conclusion can be drawn?

   (a) 1010 (4 bits), (b) 111010 (6 bits), (c) FA_{16} (8 bits), (d) $3FA_{16}$ (10 bits).

6. For each of the following pairs (A, B) of numbers, calculate A + B, A - B, -A + B, and -A - B, assuming an 8-bit two's complement number system. Check the results by decimal arithmetic. Explain any unusual results.

   (a) 01010101, 00001010 ($+95_{10}$, $+75_{10}$, -75_{10}, -95_{10})

   (b) 01101011, 00101010 (Overflow, $+65_{10}$, -65_{10}, Overflow)

   (c) 11101010, 00101111 ($+25_{10}$, -69_{10}, $+69_{10}$, -25_{10})

   (d) 10000000, 01111111 (-1_{10}, Overflow, Can't be obtained, Can't be obtained)

   Discussion: How can overflow be detected when adding 2's complement numbers?

7. Write the 8-bit signed-magnitude, two's complement and ones' complement representation for each decimal number: +82, +120, -42, and -111.

   Ans:

Decimal	+82	+120	-42	-111
Signed-magnitude	01010010	01111000	10101010	11101111
Two's-complement	01010010	01111000	11010110	10010001
Ones'-complement	01010010	01111000	11010101	10010000

8. Encode each of the following decimal numbers in BCD.

   (a) 39 (b) 625 (c) 1950 (d) 94704

   Discussion: Is a BCD number a binary or a decimal number?

9. A register (to be studied in later part of the course) of n bits in size is a memory for storing an n-bit data. When a 12-bit register stores 100010010111, what is its content if it represents:

   (a) three decimal digits in BCD; Ans: 897

   (b) an unsigned binary number; Ans: 2199

   (c) a 2's complement number. Ans: -1897

* More exercises can be found on page 74 of the textbook.

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

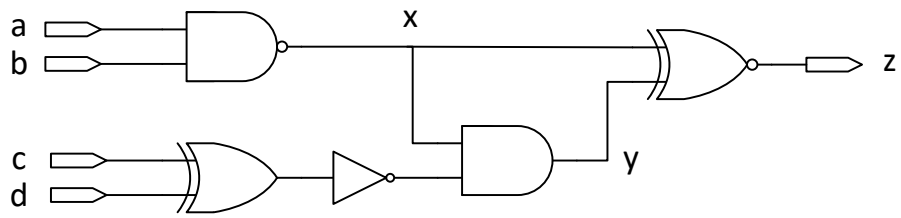
TUTORIAL 2

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1. The Stub Series Terminated Low Voltage Logic (SSTV) family, used for SDRAM modules, defines a LOW signal to be in the range 0.0–0.7 V and a HIGH signal to be in the range 1.7–2.5 V. Under a positive-logic convention, indicate the logic value associated with each of the following signal levels:

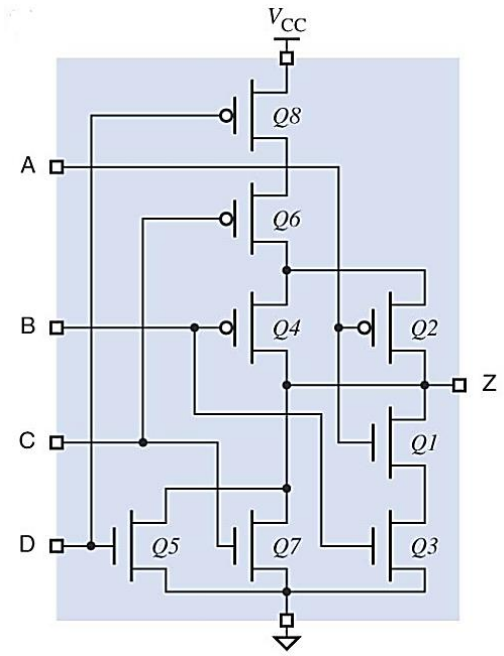
   (a) 0.0 V (b) 0.7 V (c) 1.7 V (d) −0.6 V (e) 1.6 V (f) −2.0 V (g) 2.5 V (h) 3.3 V.

2. Obtain the truth table of the logic circuit below.

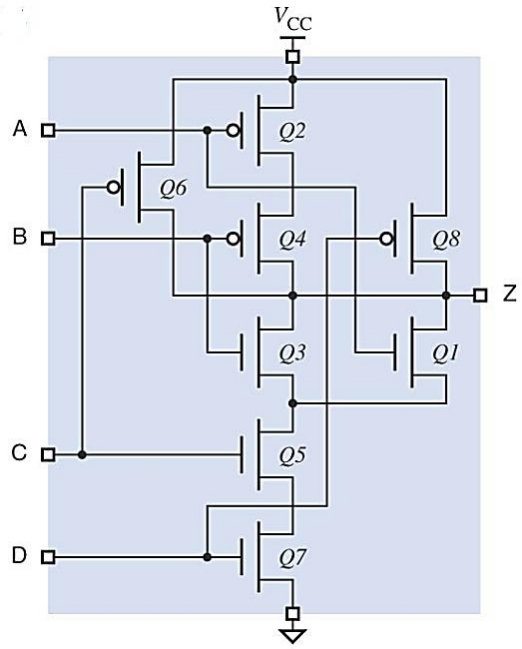


3. The circuit below is a type of CMOS AND-OR-INVERT gate. Write a function table for this circuit in the style of Figure 3-15(b) (see Textbook page 91, or Lecture Slide 74 "CMOS 2-input NOR Gate"), and a corresponding logic diagram using AND and OR gates and inverters.

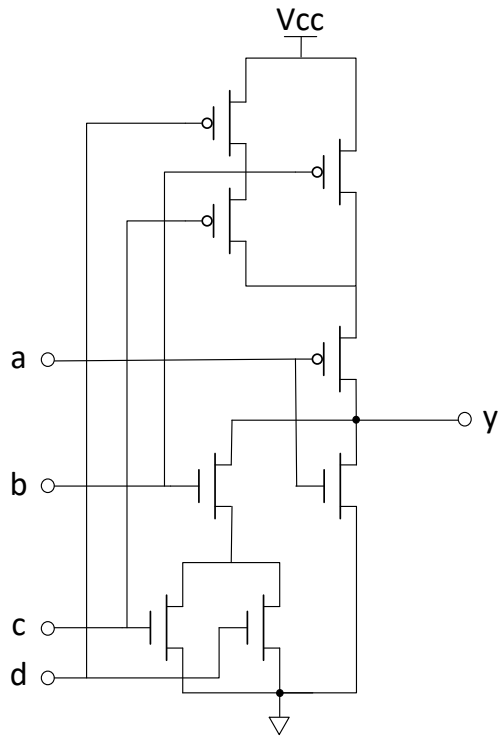
   Ans: See Lecture Slide 84



4. The circuit below is a type of CMOS OR-AND-INVERT gate. Write a function table for this circuit in the same style of Question 3, and a corresponding logic diagram using AND and OR gates and inverters.



5. Derive the logic function of the following CMOS circuit.



6. Using minimum number of transistors, design a CMOS circuit for the logic function $y = \overline{a \cdot (b + c \cdot d)}$.

7. Discuss how the PMOS and NMOS transistors are connected in Questions 3 to 6. Hence, suggest a general rule for connecting the PMOS transistors and NMOS transistors in the design of CMOS circuits.

* More exercises can be found in the textbook.

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 3

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1. With minimum number of transistors, draw a transistor-level CMOS circuit for the logic function, $f(a,b,c,d)=\overline{(a+b\cdot\overline{c})\cdot d+\overline{a}}$. Note that you do not need to simplify the function.

2. Draw a CMOS circuit for the function $f=[a+(c+d)\cdot(b+c)]\cdot(\overline{a}+\overline{c})$.

3. Given the characteristics of the logic families 74AS (Advanced Schottky TTL) and 74ALS (Advanced Low-power Schottky TTL) in the highlighted columns of the table below,

Table I: Characteristics of gates in TTL families

		Family				
Description	***Symbol***	***74S***	***74LS***	***74AS***	***74ALS***	***74F***
Maximum propagation delay (ns)		3	9	1.7	4	3
Power consumption per gate (mW)		19	2	8	1.2	4
Speed-power product (pJ)		57	18	13.6	4.8	12
LOW-level input voltage (V)	V_{ILmax}	0.8	0.8	0.8	0.8	0.8
LOW-level output voltage (V)	V_{OLmax}	0.5	0.5	0.5	0.5	0.5
HIGH-level input voltage (V)	V_{IHmin}	2.0	2.0	2.0	2.0	2.0
HIGH-level output voltage (V)	V_{OHmin}	2.7	2.7	2.7	2.7	2.7
LOW-level input current (mA)	I_{ILmax}	−2.0	−0.4	−0.5	−0.2	−0.6
LOW-level output current (mA)	I_{OLmax}	20	8	20	8	20
HIGH-level input current (μA)	I_{IHmax}	50	20	20	20	20
HIGH-level output current (μA)	I_{OHmax}	−1000	−400	−2000	−400	−1000

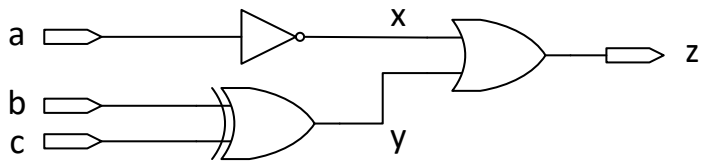
(a) Compute the fanout for a 74AS gate to drive other gates in the same family;
(b) Compute the fanout for a 74AS gate to drive gates in the 74ALS family;
(c) Compute the fanout for a 74ALS gate to drive gates in the 74AS family;
(d) Compute the noise margins of the 74AS logic family.

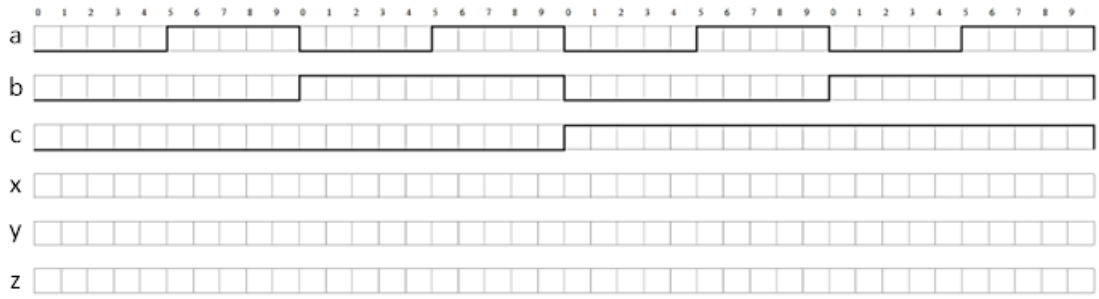
4. A 74AC/ACT gate is driving forty 74AS gates. In addition to the forty gates, the driver is required to drive some 74ALS gates. Calculate the maximum possible number of 74ALS gates that can be connected. The input and output specifications of the TTL and CMOS IC families are given in the Specification Table shown below.

Table II: Specifications of TTL and CMOS IC families

Parameter	TTL 74AS00	TTL 74ALS00	CMOS 74AC	CMOS 74ACT
V_{IH} (V)	2	2	3.85	2.0
V_{IL} (V)	0.8	0.8	1.35	0.8
V_{OH} (V)	3	3	3.76	3.76
V_{OL} (V)	0.5	0.5	0.37	0.37
I_{IH} (µA)	20	20	1	1
I_{IL} (µA)	–500	–100	–1	–1
I_{OH} (mA)	–2	–0.4	–24	–24
I_{OL} (mA)	20	8	24	24

5. Given the logic circuit below, complete the waveforms of x, y and z in the figure on the next page, assuming propagation delay of 2 ns for all gates.





Additional Home Exercise

1. Given the logic function, $Y = \left(A \cdot \overline{B} + C\right) \cdot \overline{D} \cdot E + \overline{A} \cdot \overline{C}$, draw its CMOS circuit. Note that you do not need to manipulate or attempt to simplify the function.

* More exercises can be found in the textbook.

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 4

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1. Use Boolean algebra to simplify the following expressions:

   (a) $f(x,y,z) = y\bar{z}(\bar{z} + \bar{z}x) + (\bar{x} + \bar{z})(\bar{x}y + \bar{x}z)$

   (b) $f(A,B,C,D) = \overline{AB + \bar{A}\,\bar{D} + B\bar{D} + \bar{A}B + C\bar{D}A + \bar{A}D + CD + \bar{A}\,\bar{B}\,\bar{D}}$

   Ans: (a) $y\bar{z} + \bar{x}z$; (b) $\bar{B}A\bar{C}$

2. Express $f(x,y,z) = x\bar{y} + x\bar{z}$ as a sum of minterms and as a product of maxterms.

   Ans: $\sum m(4,5,6)$ and $\prod M(0,1,2,3,7)$

3. According to DeMorgan's theorem (break the bar and change the sign), the complement of $a \cdot b + c \cdot d$ is $\bar{a} + \bar{b} \cdot \bar{c} + \bar{d}$. Yet both functions are 1 for $abcd = 1110$. How can both a function and its complement be 1 for the same input combination? What's wrong here?

4. Prove $x \cdot y + x \cdot \bar{y} = (x + y) \cdot (x + \bar{y}) = x$.

5. Write the logic expressions for the following functions and use the relation in Question 4 to simplify them.

   (a) $f(A,B,C,D) = \sum m(0,2,5,8,9,10,13)$

   (b) $f(A,B,C,D) = \Pi M(0,2,4,5,8,11,15)$

   Ans: (a) $\bar{B}\,\bar{D} + B\bar{C}D + A\bar{B}\,\bar{C}$; (b) $(A+B+D)(A+\bar{B}+C)(B+C+D)(\bar{A}+\bar{C}+\bar{D})$

6. A two-level NOR logic circuit consists of only NOR gates connecting in such a way that the signals travel through at most 2 gates from any input to any output. A two-level NAND logic circuit consists of only NAND gates. Draw a two-level NOR circuit for $y = (a + b) \cdot (c + d)$ and a two-level NAND circuit for $z = a \cdot b + c \cdot d$.

7. Use the results from Questions 4 and 6, find a minimal two-level NAND realization for the function $f(A,B,C,D) = \sum m(0,2,8,10,14,15)$.

   Ans: $f(A,B,C,D) = \bar{B}\,\bar{D} + ABC$

8. Use the results from Questions 4 and 6, find a minimal two-level NOR realization for the function $f(A,B,C,D) = \sum m(1,2,3,5,6,7,8,9,12,14)$.

   Ans: $f(A,B,C,D) = (A+C+D)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+\bar{D})$

* More exercises can be found on page 230 of the textbook.

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 5

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1. Plot the function $f(A,B,C,D) = \sum m(1,3,5,7,8,9,10,11,15)$ on the Karnaugh map and list all the **implicants**. State the **prime implicants** and **essential prime implicants**.

2. Minimize the function $f(A,B,C,D) = \sum m(0,1,4,6,9,13,14,15)$ using the Karnaugh map. Note that there is more than one solution. How many prime implicants and essential implicants of the function?

   Ans: $f(A,B,C,D) = \overline{A}\,\overline{C}\,\overline{D} + \overline{B}\,\overline{C}D + ABD + BC\overline{D}$

3. Using K-map, expand the following functions into canonical SOP form.

   (a) $f(A,B,C,D) = A\overline{B} + \overline{A}CD + B\overline{C}\,\overline{D}$

   (b) $f(A,B,C,D) = (A+\overline{B})(C+\overline{D})(\overline{A}+C)$

   Ans: (a) $f(A,B,C,D) = \sum m(3,4,7,8,9,10,11,12)$
   $= \overline{A}\overline{B}CD + \overline{A}B\overline{C}\,\overline{D} + \overline{A}BCD + A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}D + A\overline{B}C\overline{D} + A\overline{B}CD + AB\overline{C}\,\overline{D}$

   (b) $f(A,B,C,D) = \sum m(0,2,3,10,11,14,15)$
   $= \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}C\overline{D} + \overline{A}\overline{B}CD + A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}CD + ABC\overline{D} + ABCD$

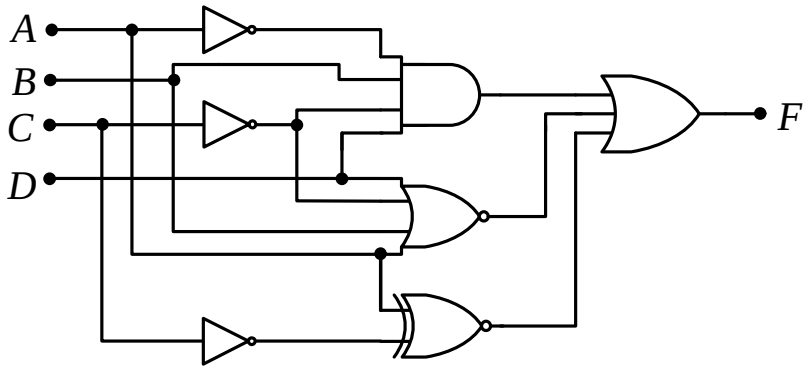
4. Given a logic function, $y(a,b,c,d) = \sum m(0,1,3,8,10,14) + x(2,4,9,11,12)$, where $x(2,4,9,11,12)$ are the don't-care terms.

   (a) Obtain the minimized sum-of-products (SOP) expression for $y(a,b,c,d)$.
   (b) Obtain the minimized product-of-sums (POS) expression for $y(a,b,c,d)$.

   Ans: (a) $y(a,b,c,d) = b' + ad'$,
   (b) $y(a,b,c,d) = (a+b')(b'+d')$
   or $y(a,b,c,d) = (a+b')(b'+d')$

5. Determine if the function $f(A,B,C,D) = ABC + AC\overline{D} + \overline{A}\,\overline{C} + B\overline{C}D$ is minimized. If not, minimize it using (a) Boolean algebra; (b) K-map.

6. Derive the function of the circuit shown below. With the help of K-map, express the function in its minimized POS form.



Ans: $F = (A + C + D)(A + B + C)(\overline{A} + \overline{C})$

Additional Home Exercise

1. Given the function, $F = \overline{x} \cdot z + \overline{\overline{w} \oplus (x \cdot \overline{y}) + w \cdot \overline{z} + w \cdot x \cdot \overline{y}}$.
   (a) Express *F* in its simplest Sum-of-Product (SOP) form and state its essential prime implicants. Hint: Use K-map.
   (b) Rewrite *F* in its simplest Product-of- Sum (POS) form.

   Ans: (a) SOP: $F = \overline{x} \cdot z + \overline{w} \cdot x \cdot \overline{y} + w \cdot y \cdot z$. All are Essential Prime Implicants
   (b) POS: $F = (x + z)(w + \overline{x} + \overline{y})(\overline{w} + \overline{x} + y)(\overline{w} + z)$

∗ More exercises can be found on page 230 of the textbook.

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 6

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1. (a) Implement functions, $X = A + B$ and $Y = A \cdot \overline{B} + \overline{B} \cdot C$ using the PLA circuit given in Figure 1(a). Note that you do not need any other logic gates.

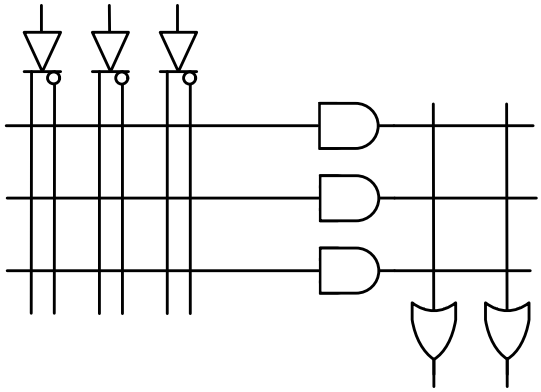


Figure 1(a)

(b) Derive the minimized SOP function of the PLA circuit given in Figure 1(b).

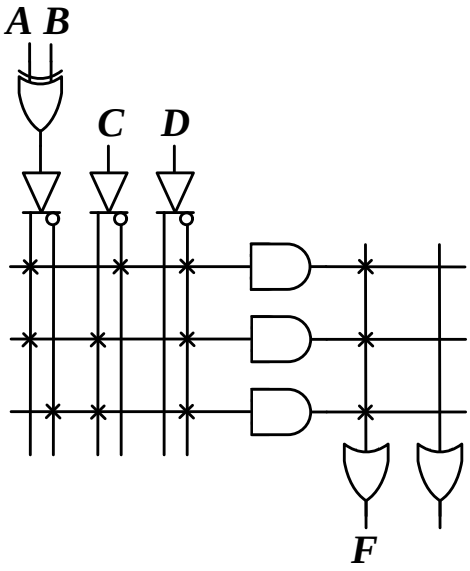


Figure 1(b)

Ans: = $\overline{A} \cdot B \cdot \overline{D} + A \cdot \overline{B} \cdot \overline{D} + C \cdot \overline{D}$

2. The function of a BCD-to-Decimal decoder is given in Figure 2(a), where $\overline{G}$ is an active low enable signal. Design such decoder using a 74LS138 3-to-8 decoder and a 74LS139 2-to-4 decoder **ONLY**. The function table of the 74LS139 is given in Figure 2(b).

Inputs					Outputs									
$\overline{G}$	A	B	C	D	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9
1	x	x	x	x	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
0	0	0	0	1	1	0	1	1	1	1	1	1	1	1
0	0	0	1	0	1	1	0	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	0	1	1	1	1	1	1
0	0	1	0	0	1	1	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1	1
0	0	1	1	1	1	1	1	1	1	1	1	0	1	1
0	1	0	0	0	1	1	1	1	1	1	1	1	0	1
0	1	0	0	1	1	1	1	1	1	1	1	1	1	0

(a) Function Table Of A BCD-To-Decimal Decoder

Input			Output			
$\overline{G}$	A	B	Y_0	Y_1	Y_2	Y_3
1	x	x	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

(b) Function Table Of 74LS139

Figure 2

3. Implement the logic function $Y = \left(A + B\overline{C}\right)\left(B + \overline{A}\,\overline{C}\right)$ using:

   (a) A 74153 4-to-1 MUX, the function table of which is given in Figure 3.

   (b) A 74LS138 3-to-8 decoder with any necessary logic gates.

Input			Output
S_1	S_0	$\overline{G}$	Y
x	x	1	0
0	0	0	C_0
0	1	0	C_1
1	0	0	C_2
1	1	0	C_3

Figure 3

4. Express the logic function F(a, b, c) of Figure 4 in the minterm and maxterm forms, where the module is a 74153 4-to-1 MUX.

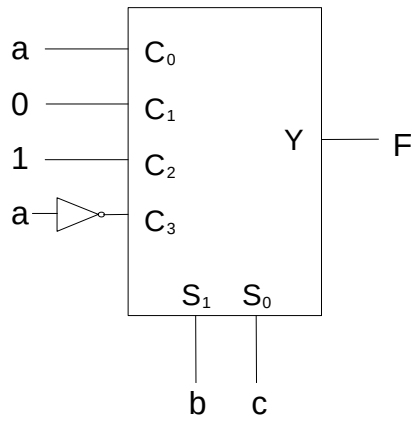


Figure 4

5. Design a 2-bit 2's complement number converter using the active-low 2-to-4 decoder and 4-to-2 encoder given below.

Function table (a): 2-to-4 decoder

Input			Output			
$\overline{G}$	A	B	Y_0	Y_1	Y_2	Y_3
1	x	x	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

Function table (b): 4-to-2 encoder

Input				Output	
A_0	A_1	A_2	A_3	Y_1	Y_0
0	1	1	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	0	1	1

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING

ACADEMIC YEAR 2026 – 2027
SEMESTER 1

IE2104 DIGITAL ELECTRONICS

TUTORIAL 7

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1. Determine the typical propagation delay from IN to OUT of the circuit in Figure 1 for both LOW-to-HIGH and HIGH-to-LOW transition, using the time information given in the table below.

(Ans: 53ns, 57ns)

	74HCT		74AHCT				74LS			
	Typical	Maximum	Typical		Maximum		Typical		Maximum	
Part Number	t_{pLH}, t_{pHL}	t_{pLH}, t_{pHL}	t_{pLH}	t_{pHL}	t_{pLH}	t_{pHL}	t_{pLH}	t_{pHL}	t_{pLH}	t_{pHL}
'00, '10	11	35	5.5	5.5	9.0	9.0	9	10	15	15
'02	9	29	3.4	4.5	8.5	8.5	10	10	15	15
'04	11	35	5.5	5.5	8.5	8.5	9	10	15	15
'08, '11	11	35	5.5	5.5	9.0	9.0	8	10	15	20
'14	16	48	5.5	5.5	9.0	9.0	15	15	22	22
'20	11	35					9	10	15	15
'21	11	35					8	10	15	20
'27	9	29	5.6	5.6	9.0	9.0	10	10	15	15
'30	11	35					8	13	15	20
'32	9	30	5.3	5.3	8.5	8.5	14	14	22	22
'86 (2 levels)	13	40	5.5	5.5	10	10	12	10	23	17
'86 (3 levels)	13	40	5.5	5.5	10	10	20	13	30	22

From *Digital Design*: *Principles and Practices*, Fourth Edition, John F. Wakerly, ISBN 0-13-186389-4.
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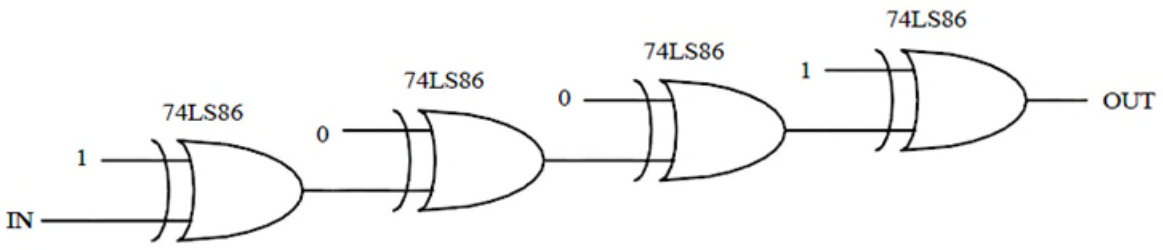


Figure 1

2. Design a binary comparator circuit to compare two 8-bit numbers using exclusive-OR gates and a NOR gate. The comparator outputs a logic 1 when the two numbers are equal. Analyze the circuit when the two inputs are A = 1101 1001 and B = 1101 1001 and when A = 1010 1110 and B = 1110 1110.

3. Design a comparator for comparing two 5-bit numbers in 2's complement form using the 7485 4-bit magnitude comparator below and any necessary basic logic gates.

Input Numbers				**Cascading Inputs**			**Outputs**		
A_3B_3	A_2B_2	A_1B_1	A_0B_0	$I_{A>B}$	$I_{A<B}$	$I_{A=B}$	$Y_{A>B}$	$Y_{A<B}$	$Y_{A=B}$
$\mathbf{A_3>B_3}$	X	X	X	X	X	X	**H**	L	L
$\mathbf{A_3<B_3}$	X	X	X	X	X	X	L	**H**	L
$A_3=B_3$	$\mathbf{A_2>B_2}$	X	X	X	X	X	**H**	L	L
$A_3=B_3$	$\mathbf{A_2<B_2}$	X	X	X	X	X	L	**H**	L
$A_3=B_3$	$A_2=B_2$	$\mathbf{A_1>B_1}$	X	X	X	X	**H**	L	L
$A_3=B_3$	$A_2=B_2$	$\mathbf{A_1<B_1}$	X	X	X	X	L	**H**	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$\mathbf{A_0>B_0}$	X	X	X	**H**	L	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$\mathbf{A_0<B_0}$	X	X	X	L	**H**	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	**H**	L	L	**H**	L	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	L	**H**	L	L	**H**	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	X	X	**H**	L	L	**H**
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	L	L	L	H	H	L
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	H	H	L	L	L	L

4. Using four half-adders to design a 4-bit incrementer (a circuit that adds one to a 4-bit binary number), and using four full-adders to design a 4-bit decrementer (a circuit that subtracts 1 from a 4-bit binary number).